Transformations, Compositions and Inverses: A Synthesis Review

Transforming a graph, composing two functions, and inverting the result

Directions

- Problems 1–3 are a warm-up, one on each of the three tools. After that the tools are mixed, so decide which one the problem is asking for before you start writing.
- Give exact answers; leave fractions as fractions.
- Where a domain restriction is stated, use it — it decides a sign.

1. Let $f(x) = x^2$ and $g(x) = f(x-2) + 3$. Find $g(5)$.

Answer: ________________

2. Let $f(x) = 2x + 1$ and $g(x) = x^2$. Find $(f \circ g)(3)$.

Answer: ________________

3. Let $f(x) = 5x - 3$. Find $f^{-1}(12)$.

Answer: ________________

4. Let $f(x) = \sqrt{x}$ and $g(x) = -2f(x+1)$. Find $g(8)$.

Answer: __________________

5. Let $f(x) = x^2 - 1$ and $g(x) = x + 4$. Write $(f \circ g)(x)$ as a simplified polynomial.

Answer: __________________

6. Let $f(x) = \dfrac{x-7}{2}$. Write the rule for f^{-1}, then find $f^{-1}(4)$.

Answer: __________________

7. Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 1$. Simplify $(f \circ g)(x)$, then solve $(f \circ g)(x) = 3$.

Answer: ________________

8. Let $f(x) = x^2 - 17$ and $g(x) = 3f(x-2) + 5$.

(a) Find $g(6)$.

Answer: ________________

(b) Describe, in order, the transformations that take $y = f(x)$ to $y = g(x)$.

Answer: ________________

9. Let $f(x) = (x-3)^2 + 1$ with the domain restricted to $x \geq 3$. Find $f^{-1}(17)$.

Answer: ________________

10. Let $f(x) = \dfrac{1}{x+2}$ and $g(x) = \dfrac{1}{x} - 2$.

(a) Simplify $(f \circ g)(x)$.

Answer: ____________________

(b) Explain what your answer says about the relationship between f and g.

Answer: ____________________

11. Let $f(x) = 2x + 1$ and $h(x) = f^{-1}(x - 4)$. Find $h(11)$.

Answer: ____________________

12. Let $f(x) = 3x - 5$ and $g(x) = x^2$ with the domain of g restricted to $x \geq 0$.

(a) Find $(g \circ f)^{-1}(9)$.

Answer: ____________________

(b) Explain why $(g \circ f)^{-1}$ equals $f^{-1} \circ g^{-1}$ rather than $g^{-1} \circ f^{-1}$.

Answer: ____________________